

LeetCode 802: Eventual Safe States

Atlassian

Adobe

DeHaat

Visual Logic & Notes

01 Basic idea ye rkhenge ki: terminal node wo node he jiska outdegree 0. To initially queue me saare terminal node daalenge.

02 Ab peeche traverse krne ke liye: revadj banayenge (reverse adjacency list).

03 Ek baar terminal node process ho gya to apne se peeche waalo ki outdegree kam krega.

04 Agar uske peeche waale ki outdegree 0 hogyi, iska matlab usse jitne bhi edge nikal rhe the wo sirf terminal nodes tak jaa rhe the → **push to answer**.

05 Agar outdegree 0 nhi hui to terminal node ke alawa bhi kahi gaand mara rha (meaning it has edges leading to unsafe paths).

| Source Code (C++)

```

class Solution {
public:
    vector<int> eventualSafeNodes(vector<vector<int>>& graph) {
        //basic idea ye rkhenge ki
        //terminal node wo node he jiska outdegree 0
        //to intially queue me saare terminal node
        //ab peeche traverse krne ke liye revadj
        //ek baar terminal node process ho gya to apne se peeche waalo ki outdeg
        //agar uske peeche waale ki outdegree 0 hogyi
        //iska matlab usse jitne bhi edge nikal rhe the wo sirf terminal nodes t
        //push to answer
        //agar outdegree 0 nhi hui to terminal node ke alawa bhi kahi gaand mara

        int n=graph.size();
        vector<vector<int>> revadj(n);
        vector<int> outdeg(n);

        for(int u=0;u<n;u++){
            outdeg[u]=graph[u].size();
            for(auto v:graph[u]){
                revadj[v].push_back(u);
            }
        }

        queue<int> q;
        vector<int> ans;
        for(int i=0;i<n;i++){
            if(outdeg[i]==0){
                q.push(i);
            }
        }

        while(!q.empty()){
            int curr=q.front();
            ans.push_back(curr);
            q.pop();

            for(auto &it:revadj[curr]){
                outdeg[it]--;
                if(outdeg[it]==0){
                    q.push(it);
                }
            }
        }

        sort(ans.begin(),ans.end());
        return ans;
    }
};

```

```
}  
};
```